Yongye(Felix) Hu

510-277-7666 | hyy.felix@gmail.com | felix-hyy | felix-hu.space

EDUCATION

University of California, Berkeley **Aug 2024 - May 2025**
Master of Engineering | Robotics Engineering *Berkeley, CA*

Wuhan University of Science and Technology **Sep 2020 - May 2024**
Bachelor of Engineering (Honours) | Mechatronics Engineering *Wuhan, China*

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd. **Sep 2023 - Mar 2024**
Mechatronics Engineer Intern *Wuhan, China*

- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Integrated sensors including IR, microswitches, and pressure transducers to monitor system state and trigger alarms, enhancing operational reliability and safety.

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform **Sep 2024 - May 2025**
Robotics Engineer & Project Status Manager *Berkeley, CA*

- Developed a ROS2-based robotic system integrating a simulator and MPC controller, enabling precise obstacle avoidance and full trajectory completion with real-time state logging.
- Applied control system design and experimental validation to establish an OptiTrack-based real-time feedback pose control system, enabling sub-centimeter trajectory tracking and iterative controller refinement.

ROBOCON: Multi-Robot Coordination System **Sep 2022 - Jul 2023**
Integrated Mechanical Engineer *Wuhan, China*

- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.
- Modularized the robot into gimbal, power, launcher, and chassis to achieve independent functionality and decoupled operation, incorporating modular interfaces and redundancy to improve system stability and debuggability.

Lightweight, Sustainable Remote-Mining Excavator System **Jan 2021 - Jul 2023**
R&D Engineer *Wuhan, China*

- Designed a compact, low-power crane system with ESP32-based Wi-Fi remote control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100-MHz oscilloscope.

SKILLS

- **Hardware:** FDM/SLA 3D Printing, CNC, PCB, STM32CubeMX, KiCad, Keil 5
- **Design:** SolidWorks, Autodesk Fusion, Inventor, Ansys, MATLAB, AutoCAD
- **Electronics:** STM32, ESP32, Arduino, Raspberry Pi, Modbus RTU, PLC
- **Programming:** C++, Python, Linux, ROS2, MATLAB, Qt, LaTeX, HTML